

BUDDHI RAI

University of Hawai Maui College
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Education

Ph.D.	<i>Western Michigan University</i> , Kalamazoo, MI Dissertation: Modeling Transmission of Photonic Crystal Waveguide Modes Enhanced by Kerr Nonlinearity; Advisor: Dr. Arthur McGurn; Coursework GPA: 4.0/4.0	August 2012
M.S.	<i>Central Michigan University</i> , Mount Pleasant, MI Thesis: Quantum confinement of nanostructured systems; Advisor: Dr. Marco Fornari	August 2005
M.Sc.	<i>Tribhuvan University, Kirtipur</i> , Kathmandu, Nepal In lieu of Thesis: Solid-state theory	March 1994
B.Sc.	<i>Tribhuvan University, Amrit Campus</i> , Kathmandu, Nepal Majors in Physics, Chemistry, and Mathematics	March 1990

Academic Leadership, Program Development and Teaching Innovation

University of Hawaii Maui College STEM-Physics Department Kahului, HI

Associate Professor of Physics August 2022 - Present

- Developed and implemented the research experience travel-study program, featuring hands-on high-altitude science and the physics of elevated environments in action, conducted in both Hawaii and Nepal.
- Advanced inclusive, research-based learning by mentoring students across diverse STEM pathways & platforms - including Indigenous Knowledge in Engineering Program ('IKE), Undergraduate Research Experience (URE), Bridge to the Baccalaureate (B2B), Kaihuwa'a Scholarship Program (KPSII), and Physics Education Research (PER) groups - integrating culturally grounded knowledge, interdisciplinary collaboration, and hands-on inquiry to cultivate transformative educational experiences.
- Led scholarly exchanged cross-cultural learning program, and actively engaged the academic community through initiatives such as knowledge sharing guest lectures model, promoting international collaboration & cross-cultural understanding, and building global education partnerships.
Outcomes: connect students and educators in meaningful ways.

Assistant Professor of Physics August 2017 - July 2022

- Created a comprehensive ASNS program map highlighting various degree concentrations and STEM pathways available for students
- Developed physics curricula for engineering technology, electronic & computer engineering, technical automotive, and electrical & construction technology programs

Faculty STEM Physics Step-4 (October - December educational break) October 2013 - July 2017

- Developed and taught College Physics, General Physics, and Engineering Physics lectures & labs

- Taught Conceptual Physics lectures and laboratories
- Developed physical science course and/or program for non-science majors

Kalamazoo Valley Community College Department of Physics

Kalamazoo, MI

Adjunct Faculty

August 2012 - August 2013

- Taught lectures & labs for College Physics I & II courses. Used **Moodle** technology for posting classroom materials including syllabus, lectures, homework, quizzes, exams, grades, and so on.

Western Michigan University Department of Physics

Kalamazoo, MI

Assistant Course Instructor

August 2012 - August 2013

- Re-designed the instructional model for General Physics Labs
- Designed the College Physics Experiments using Video Analysis technique & Logger-Pro Vernier sensor, trained graduate student to teach this course
- Assisted web class designed for PHYS 2050 course

Doctoral Teaching Associate

August 2006 - July 2012

- Re-designed the instructional model for Introduction to Modern Physics Labs
- Taught Modern Physics, Electricity Magnetism Light, How Things Work labs
- Instructed University Physics labs using Computer with Interactive Physics Softwares
- Instructed labs/classes using the e-learning systems (**Blackboard** and **D2L**) and iClickers

Teaching Assistant Mentor

July 2009 - June 2012

- Coordinated all TAs supporting undergraduate laboratories and lectures
- Counseled to new TAs, served as liaison between graduate students and the administration
- Served as member of undergraduate Lab Development Team, developed teaching pedagogy for modern physics labs

Teaching Assistant

August 2005 - July 2006

- Taught Sky and Solar System Laboratory
- Taught General Mechanics Lab, graded tests for University Physics lecture

WMU College of Engineering and Applied Sciences

Kalamazoo, MI

Physics Instructor

June 2010 - August 2010

- Developed and taught Science Pre-First Year Physics course for the Louis Stokes Alliances for Minority Participation (LSAMP) Program funded by NSF
- Organized and led physics and engineering lab tours, incorporating mini lab projects to engage and introduce students to practical applications

Central Michigan University Department of Physics

Mount Pleasant, MI

Teaching Assistant

August 2002 - May 2005

- Led hands-on experimnts and guided student learning in Introductory Mechanics Lab
- Facilitated and instructed Electricity and Magnetism physics labs, ensuring conceptual understanding

Tribhuvan University:[†]Padma Kanya Campus,[‡]Amrit Science Campus Kathmandu, Nepal

Assistant Physics Lecturer

[†]June 1995 - August 1997, [‡]June 1995 - July 2002

- Taught Thermodynamics, Electricity, Modern Physics lectures and general labs
- Tutored students in remedial classes

Teaching Interests

- Fundamentals of Physics: mechanics, thermodynamics, electricity & magnetism, and modern physics
- Physics Education Research for laboratory courses and STEM outreach programs
- Curriculum development and teaching for Hybrid and Online lecture/laboratory courses
- College physics for pre-science, pre-engineering, non-science, and/or non-technical majors
- Physics for Electronics & Computer Engineering, Engineering Technology, and Medical Science
- Advanced level undergraduate Classical and Quantum Mechanics

Teaching Approach

- Enhancing student learning by creating interactive classroom/lab environment
- Integrating online physics laboratories (*e.g.*, PhET Simulations) with experiential learning model
- Developing & practicing evidence-based teaching including relevance, connection and enjoyment
- Emphasizing on the importance of the real-world applications and linking classroom materials to students' everyday lives
- Implementing differentiated instructional strategies including visual, auditory, participatory, computational model, and simulations. Focusing on student-centered approach to education
- Emphasizing physics curriculum appreciating and including diversity in the classroom
- Embedding classroom instructional technology, including on-line learning systems
- Teaching And Learning Utilizing Locally Available Materials (TALULAM)

University/College Service (to be updated)

- Co-ordinator of ASNS program at UH Maui College Fall 2015 - Fall 2021
- Chair of Screening Committee & DPC at UHMC &/ UH System Spring 2016 - various
- Member of Liberal Arts Executive Committee UH Maui College Spring 2015 - Spring 2016
- Member of Academic Senate UH Maui College Spring 2014 - present
- Member of Undergraduate Lab Development Team WMU Summer 2011 - Fall 2013
- Representative of WMU Physics Department in the *Graduate and Professional School Fair*, Kalamazoo, MI October 2009, October 2010, October 2011

Community Service (to be updated)

- **Ka Hikina O Ka La Mitigation program** Summer Bridge 2014, 2015, 2016, 2017 & 2018
Designed and taught hands-on projects to excite, invigorate, and engage Native Hawaiian student learning to the STEM field, UHMC, Kahului HI
- **Maui Science Olympiad Tournament** Every Spring, from 2015 to the present
Served as a Judge for the Experimental Design, UHMC, Kahului HI

- **STEAM Ahead** March 2014, March 2015, April 2016
Conducted interactive hands-on sessions, UHMC Department of Physics, Kahului HI
- **UHMC High School Counselors Workshop** September 2015
Engaged Maui county HS counselors with an intimate and interactive session, UHMC Department of Physics, Kahului HI
- **Nepali Interpretation Assignments** December 2012 - Present
Served as English-Nepali language translator, Bronson Hospital, Kalamazoo MI, & other states

Awards and Honors (to be updated)

- **2nd Annual Faculty Appreciation** Fall 2015
Recognition for dedication to UHMC and motivating students to be on top of their work!
University of Hawaii Maui College, The Student Government, Kahului, HI
- **Haym Kruglak Graduate Student Teaching Excellence Award** April 2011
Demonstrated the knowledge of physics and a sensitivity to students,
Western Michigan University, Department of Physics, Kalamazoo, MI
- **2011-12 Graduate Teaching Effectiveness Award** April 2012
Demonstrated the teaching excellence in laboratory and classroom,
Western Michigan University, The Graduate College, Kalamazoo, MI
- **Phi Kappa Phi ($\Phi K \Phi$) Academic Honor** April 2009
Ranked in the upper 10 percent of the graduate class
- **David Carley Memorial Graduate Fellowship Award** April 2006
Demonstrated outstanding scholarly achievement,
Western Michigan University, Department of Physics, Kalamazoo, MI
- **Best Teaching Award** April 2000
Demonstrated excellence in practice-based labs for high school science curriculum,
Ideal English School, Kathmandu, Nepal

Research Interests

- (*Physics Education Research during my professorship*) Developing and Implementing hands-on experimental models to enhance classroom learning and community outreach, aiming to improve physics education and student engagement through practical, inquiry-based approaches.
- (*Core Research during my MS & PhD years*) Specializing in computer simulation and modeling; designing photonic crystal waveguides and resonators; photonic crystal fiber optics; optical integrated circuits; nanoscale and electromagnetic materials; quantum confinement of low-dimensional systems.

Research Experience

Western Michigan University Department of Physics Kalamazoo, MI

Doctoral Associate August 2006 - July 2012

- Accomplished dissertation research projects on Kerr nonlinear modes in the Photonic Crystal Waveguides using Difference Equation approach.
- Implemented computer simulation techniques for modelling one- and two-dimensional photonic crystal waveguides and resonators.

- Investigated spin properties on nanostructured and half-metallic systems
- Investigated electronic structures of Quantum Wells and Superlattices using Mathematica programming tools

Publications and Collaborations (to be updated)

- Buddhi Rai and Arthur McGurn, Multiplicity of transmission coefficients in photonic crystal and split ring resonator waveguides with Kerr nonlinear impurities, Phys. Rev. B 91, 085113 – Published 19 February 2015
- Buddhi Rai and Manuel Bautista, proposed a final draft of Introduction to Elementary Physics Lab Manual to the Department of Physics, Western Michigan University, Kalamazoo, MI (Date: July 1, 2013)
- Kandiah Balachandran and Buddhi Rai, Influence of Fluids on Compressional to Shear Wave Conversion in Reservoirs, Kalamazoo Valley Community College, Kalamazoo, MI (Working paper. Data analysis using OpendTect seismic interpretation software system)
- Buddhi Rai, Modeling of a Photonic Crystal Waveguide Mode with the FDTD Method, The Hilltop Review, Vol. 4, Spring 2011, Western Michigan University
Link: <http://www.wmich.edu/gsac/publications.html>, Kalamazoo, MI
- Manuel Bautista, Alvin Rosenthal, Christopher A. Hoffman, Buddhi Rai, T. M. Elkafrawy, Introduction to Modern Physics Lab Manual, Department of Physics, Western Michigan University, Kalamazoo, MI (Date: August 21, 2011)

Selected Presentations, Seminars and Conferences (to be updated)

- **2024 HISSI Conference - Building Dreams, Strengthening Communities** March 8 2024
Interactive workshop presentation on “Journeys of Discovery: Faculty-Led Travel Programs that Impact on Students and Educators,” Hawai’i Convention Center, Honolulu; Co-PI: Liana Horovitz, Prof. of History; *UHCC travel grant support*
- **2023 HISSI - Shared Purpose in a Post-Pandemic World** March 3 2023
Presented Original Research Data on “Are we providing quality education after Covid-19?” Hawai’i Convention Center, Honolulu; *UHCC travel grant support*
- **2022 Excellence in Education Day** March 4 2022
Workshop presentation on “Virtual Physics Labs vs. Traditional Hands-On Labs: Literature Review and Curriculum Work,” 2022 Excellence in Education Day, mālama kekahi i kekahi, Hawai’i; *LSAMP B2B NSF grant support*
- **Hawai’i Student Success Institute 2019** March 2-3 2019
Workshop presentation on “Teaching and Learning Physics/Physical Sciences Utilizing Locally Available Materials (TALULAM),” Honolulu;
- **2017 Hawai’i Student Success Institute Student Success Pathways** March 3-4
Workshop presentation on “Teaching Physical Sciences Utilizing a Non-Mathematical Approach,” Honolulu;

- 2016 Hawai'i STEM Summit** September 30 2016
Participated in group activity & seminar agendas: The Current and Future State of Hawaii STEM Workforce, Aligning STEM Education Pathways, etc., Ko'olau, Ballroom, Kāneohe, Hawaii; funded by USA Funds
- Hawai'i National Great Teachers Seminars** August 7-12 2016
Presented/shared teaching innovation and problem, Kilauea Military Camp, Hawaii Volcanoes National Park, Hawaii; funded by Perkins grant
- Hawai'i Strategy Institute Meeting** March 6-7 2015; March 4-5 2016
Participated and shared the best teaching practices, and accelerated student success, Honolulu, Hawaii; funded by UH travel grant
- American Physical Society Ohio-Section Spring 2012 Meeting** April 13-14 2012
*Contributed poster presentation: Modeling of the Kerr-nonlinear Modes in Photonic Crystal Waveguide with the FDTD Method, Columbus, OH; received **APS Travel Grant***
- American Physical Society March Meeting 2012** February 27-March 2 2012
*Contributed paper presentation: Kerr Nonlinear Modes in Photonic Crystal Waveguide with Off-channel Features, Boston, MA; received **Western Michigan University Graduate Student Research and Travel Grant 2012 Award** for this work*
- 2nd Annual Multidisciplinary Science Forum** October 2011
Contributed poster presentation: Simulation of the nonlinear modes in photonic crystal waveguides, Japan Society for the Promotion of Science, The Soga Japan Center, conference held at Western Michigan University, Kalamazoo, MI

Association Memberships

- American Federation of Teachers membership** April 2008 - present
 Localized with the University of Hawaii Professional Assembly, Hawaii, now, and it started from the Teaching Assistants Union of Western Michigan University, Michigan
- American Physical Society membership** August 2004 - present
- Phi Kappa Phi ($\Phi K \Phi$)** April 2009 - present
- Ohio Section of APS Condensed Matter Division membership** August 2004 - present
- Nepal Physical Society** Lifetime Membership since 1995

References

- Sean Calder, Professor & Chair
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arthur.mcgurn@wmich.edu Ph.D. dissertation advisor
- Dr. Manuel Bautista, Professor
Department of Physics Western Michigan University
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(269) 387-5360 office;
manuel.bautista@wmich.edu course supervisor (2011 - 2013)
- Dr. Kandiah Balachandran [BALA], Professor
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(269) 488-4752 office; kbalachandra@kvcc.edu lead professor (2012 - 2013)
- Dr. Thomas Gorczyca, Professor
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